

MATH120-24S2 Discrete Mathematics

Assignment 3

1. Use induction to prove that, for all $n \in \mathbb{Z}^+$, $1 + 4 + 16 + 64 + \dots + 4^{n-1} = \frac{4^n - 1}{3}$

Solution:

Base step. If $n = 1$, then $1 = \frac{4 - 1}{3}$, so the result holds for $n = 1$.

Induction step. Now assume that the result holds when $n = k$, that is, assume

$$1 + 4 + 16 + 64 + \dots + 4^{k-1} = \frac{4^k - 1}{3}$$

We next show that the result holds when $n = k + 1$.

By the induction hypothesis,

$$\begin{aligned} 1 + 4 + 16 + 64 + \dots + 4^{k-1} + 4^k &= \frac{4^k - 1}{3} + 4^k \\ &= \frac{4^k - 1 + 3 \cdot 4^k}{3} \\ &= \frac{4 \cdot 4^k - 1}{3} \\ &= \frac{4^{k+1} - 1}{3} \end{aligned}$$

Hence by the principle mathematical induction,

$$1 + 4 + 16 + 64 + \dots + 4^{n-1} = \frac{4^n - 1}{3}$$

for all $n \in \mathbb{Z}^+$.

2. Suppose that a_n is defined by setting $a_1 = 4$, $a_2 = 12$, and $a_n = 4a_{n-1} - 4a_{n-2}$, where $n \geq 3$. Using strong induction, prove that $a_n = 2^n(n + 1)$ for all $n \in \mathbb{Z}^+$.

Solution:

Let $S(n)$ be the statement that $a_n = 2^n(n + 1)$.

Base step. When $n = 1$, we have $4 = 2(1 + 1)$. When $n = 2$ we have $12 = 2^2(2 + 1)$, so we have that both $S(1)$ and $S(2)$ are true.

Induction step. Now assume that $S(m)$ is true for all integers m with $1 \leq m \leq k$. We need to show that $S(k + 1)$ is true.

We have that

$$a_{k+1} = 4 \cdot a_k - 4 \cdot a_{k-1}.$$

By the induction hypothesis, both $S(k)$ and $S(k - 1)$ are true. This means that $a_k = 2^k(k + 1)$ and $a_{k-1} = 2^{k-1}k$.

Hence, we have that

$$\begin{aligned} a_{k+1} &= 4 \cdot 2^k(k + 1) - 4 \cdot 2^{k-1}(k) \\ &= 2^{k+1}(2k + 2) - 2^{k+1}(k) \\ &= 2^{k+1}(k + 2). \end{aligned}$$

That is, $S(k + 1)$ is true. The result now follows by strong induction.

3. Consider the following statement: Let a, b be integers. If $a + b \geq 9$ then $a \geq 5$ or $b \geq 5$.

We asked ChatGPT to provide a proof by contradiction of this statement and received this:

Sure! Let's prove this statement by contradiction.

Statement: If $a + b \geq 9$, then $a \geq 5$ or $b \geq 5$.

Proof by Contradiction:

Assume the negation of the statement is true: $a + b \geq 9$ and $a < 5$ and $b < 5$.

Since $a + b \geq 9$, we have two cases:

- 1. Case 1: $a < 5$ and $b < 5$. If both a and b are less than 5, their sum can never be greater than or equal to 9. This contradicts the assumption $a + b \geq 9$.*
- 2. Case 2: $a < 5$ and $b = 5$ (or $a = 5$ and $b < 5$). Without loss of generality, let's assume $b = 5$ and $a < 5$. Then $a + b = a + 5 \geq 9$. Subtracting 5 from both sides, we get $a \geq 4$. But this contradicts the assumption that $a < 5$.*

Hence, in both cases, our assumption leads to a contradiction. Therefore, the original statement is proven true by contradiction. If $a + b \geq 9$, then $a \geq 5$ or $b \geq 5$.

We then asked to provide a proof by contraposition.

Of course! Let's prove the statement by contraposition.

Statement: If $a + b \geq 9$, then $a \geq 5$ or $b \geq 5$.

Proof by Contraposition:

The contrapositive of the statement is: If $a < 5$ and $b < 5$, then $a + b < 9$.

Assume $a < 5$ and $b < 5$.

Since both a and b are less than 5, their sum, $a + b$, is less than 10. Specifically, $a + b < 5 + 5 = 10$, which implies $a + b < 9$.

Therefore, by contraposition, if $a + b \geq 9$, then $a \geq 5$ or $b \geq 5$.

- (a) Point out at least one major mathematical error in each of the proofs.
- (b) Provide a correct proof of the statement (using your method of choice).

Solution:

- (a) The proof by contradiction only needs to cover Case 1 since we assume that $a + b \geq 9$ and $a < 5$ and $b < 5$. In case 1, it is said that if both a and b are less than 5 their sum can never be at least 9. This is only correct if we use that a and b are integers which ChatGPT failed to mention. More disturbingly, in case 2, ChatGPT produces nonsense: it says that $b = 5$ and $a < 5$ implies $a + b \geq 9$ which is clearly false. In the proof by contraposition it is claimed that $a + b < 10$ implies that $a + b < 9$, also clearly a false statement.
- (b) We will provide a proof by contradiction. First note that the negation of $a \geq 5$, $a < 5$, for an integer says that $a \leq 4$ (this is the step that ChatGPT failed to make); Furthermore, if not ($a \geq 5$ or $b \geq 5$), then $a \leq 4$ and $b \leq 4$. So we need to show that if $a + b \geq 9$, and $a \leq 4$, and $b \leq 4$, we find a contradiction. So suppose that $a + b \geq 9$, $a \leq 4$ and $b \leq 4$. Then clearly $a + b \leq 8$, which is a contradiction since $a + b \geq 9$.